
Chapter 3

Cohomology

Cohomology is an algebraic variant of homology, the result of a simple dualization in the definition. Not surprisingly, the cohomology groups $H^i(X)$ satisfy axioms much like the axioms for homology, except that induced homomorphisms go in the opposite direction as a result of the dualization. The basic distinction between homology and cohomology is thus that cohomology groups are contravariant functors while homology groups are covariant. In terms of intrinsic information, however, there is not a big difference between homology groups and cohomology groups. The homology groups of a space determine its cohomology groups, and the converse holds at least when the homology groups are finitely generated.

What is a little surprising is that contravariance leads to extra structure in cohomology. This first appears in a natural product, called *cup product*, which makes the cohomology groups of a space into a ring. This is an extremely useful piece of additional structure, and much of this chapter is devoted to studying cup products, which are considerably more subtle than the additive structure of cohomology.

How does contravariance lead to a product in cohomology that is not present in homology? Actually there is a natural product in homology, but it takes the somewhat different form of a map $H_i(X) \times H_j(Y) \rightarrow H_{i+j}(X \times Y)$ called the *cross product*. If both X and Y are CW complexes, this cross product in homology is induced from a map of cellular chains sending a pair (e^i, e^j) consisting of a cell of X and a cell of Y to the product cell $e^i \times e^j$ in $X \times Y$. The details of the construction are described in §3.B. Taking $X = Y$, we thus have the first half of a hypothetical product

$$H_i(X) \times H_j(X) \rightarrow H_{i+j}(X \times X) \rightarrow H_{i+j}(X)$$

The difficulty is in defining the second map. The natural thing would be for this to be induced by a map $X \times X \rightarrow X$. The multiplication map in a topological group, or more generally an H-space, is such a map, and the resulting *Pontryagin product* can be quite useful when studying these spaces, as we show in §3.C. But for general X , the only

natural maps $X \times X \rightarrow X$ are the projections onto one of the factors, and since these projections collapse the other factor to a point, the resulting product in homology is rather trivial.

With cohomology, however, the situation is better. One still has a cross product $H^i(X) \times H^j(Y) \rightarrow H^{i+j}(X \times Y)$ constructed in much the same way as in homology, so one can again take $X = Y$ and get the first half of a product

$$H^i(X) \times H^j(X) \rightarrow H^{i+j}(X \times X) \rightarrow H^{i+j}(X)$$

But now by contravariance the second map would be induced by a map $X \rightarrow X \times X$, and there is an obvious candidate for this map, the diagonal map $\Delta(x) = (x, x)$. This turns out to work very nicely, giving a well-behaved product in cohomology, the cup product.

Another sort of extra structure in cohomology whose existence is traceable to contravariance is provided by cohomology operations. These make the cohomology groups of a space into a module over a certain rather complicated ring. Cohomology operations lie at a depth somewhat greater than the cup product structure, so we defer their study to §4.L.

The extra layer of algebra in cohomology arising from the dualization in its definition may seem at first to be separating it further from topology, but there are many topological situations where cohomology arises quite naturally. One of these is Poincaré duality, the topic of the third section of this chapter. Another is obstruction theory, covered in §4.3. Characteristic classes in vector bundle theory (see [Milnor & Stasheff 1974] or [VBKT]) provide a further instance.

From the viewpoint of homotopy theory, cohomology is in some ways more basic than homology. As we shall see in §4.3, cohomology has a description in terms of homotopy classes of maps that is very similar to, and in a certain sense dual to, the definition of homotopy groups. There is an analog of this for homology, described in §4.F, but the construction is more complicated.

The Idea of Cohomology

Let us look at a few low-dimensional examples to get an idea of how one might be led naturally to consider cohomology groups, and to see what properties of a space they might be measuring. For the sake of simplicity we consider simplicial cohomology of Δ -complexes, rather than singular cohomology of more general spaces.

Taking the simplest case first, let X be a 1-dimensional Δ -complex, or in other words an oriented graph. For a fixed abelian group G , the set of all functions from vertices of X to G also forms an abelian group, which we denote by $\Delta^0(X; G)$. Similarly the set of all functions assigning an element of G to each edge of X forms an abelian group $\Delta^1(X; G)$. We will be interested in the homomorphism $\delta: \Delta^0(X; G) \rightarrow \Delta^1(X; G)$ sending $\varphi \in \Delta^0(X; G)$ to the function $\delta\varphi \in \Delta^1(X; G)$ whose value on an oriented

edge $[v_0, v_1]$ is the difference $\varphi(v_1) - \varphi(v_0)$. For example, X might be the graph formed by a system of trails on a mountain, with vertices at the junctions between trails. The function φ could then assign to each junction its elevation above sea level, in which case $\delta\varphi$ would measure the net change in elevation along the trail from one junction to the next. Or X might represent a simple electrical circuit with φ measuring voltages at the connection points, the vertices, and $\delta\varphi$ measuring changes in voltage across the components of the circuit, represented by edges.

Regarding the map $\delta: \Delta^0(X; G) \rightarrow \Delta^1(X; G)$ as a chain complex with 0's before and after these two terms, the homology groups of this chain complex are by definition the simplicial cohomology groups of X , namely $H^0(X; G) = \text{Ker } \delta \subset \Delta^0(X; G)$ and $H^1(X; G) = \Delta^1(X; G) / \text{Im } \delta$. For simplicity we are using here the same notation as will be used for singular cohomology later in the chapter, in anticipation of the theorem that the two theories coincide for Δ -complexes, as we show in §3.1.

The group $H^0(X; G)$ is easy to describe explicitly. A function $\varphi \in \Delta^0(X; G)$ has $\delta\varphi = 0$ iff φ takes the same value at both ends of each edge of X . This is equivalent to saying that φ is constant on each component of X . So $H^0(X; G)$ is the group of all functions from the set of components of X to G . This is a direct product of copies of G , one for each component of X .

The cohomology group $H^1(X; G) = \Delta^1(X; G) / \text{Im } \delta$ will be trivial iff the equation $\delta\varphi = \psi$ has a solution $\varphi \in \Delta^0(X; G)$ for each $\psi \in \Delta^1(X; G)$. Solving this equation means deciding whether specifying the change in φ across each edge of X determines an actual function $\varphi \in \Delta^0(X; G)$. This is rather like the calculus problem of finding a function having a specified derivative, with the difference operator δ playing the role of differentiation. As in calculus, if a solution of $\delta\varphi = \psi$ exists, it will be unique up to adding an element of the kernel of δ , that is, a function that is constant on each component of X .

The equation $\delta\varphi = \psi$ is always solvable if X is a tree since if we choose arbitrarily a value for φ at a basepoint vertex v_0 , then if the change in φ across each edge of X is specified, this uniquely determines the value of φ at every other vertex v by induction along the unique path from v_0 to v in the tree. When X is not a tree, we first choose a maximal tree in each component of X . Then, since every vertex lies in one of these maximal trees, the values of ψ on the edges of the maximal trees determine φ uniquely up to a constant on each component of X . But in order for the equation $\delta\varphi = \psi$ to hold, the value of ψ on each edge not in any of the maximal trees must equal the difference in the already-determined values of φ at the two ends of the edge. This condition need not be satisfied since ψ can have arbitrary values on these edges. Thus we see that the cohomology group $H^1(X; G)$ is a direct product of copies of the group G , one copy for each edge of X not in one of the chosen maximal trees. This can be compared with the homology group $H_1(X; G)$ which consists of a direct *sum* of copies of G , one for each edge of X not in one of the maximal trees.

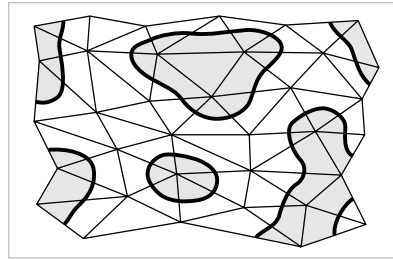
Note that the relation between $H^1(X; G)$ and $H_1(X; G)$ is the same as the relation between $H^0(X; G)$ and $H_0(X; G)$, with $H^0(X; G)$ being a direct product of copies of G and $H_0(X; G)$ a direct sum, with one copy for each component of X in either case.

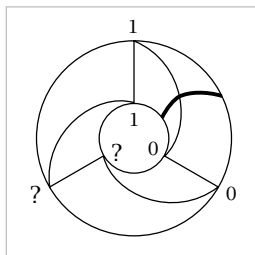
Now let us move up a dimension, taking X to be a 2-dimensional Δ -complex. Define $\Delta^0(X; G)$ and $\Delta^1(X; G)$ as before, as functions from vertices and edges of X to the abelian group G , and define $\Delta^2(X; G)$ to be the functions from 2-simplices of X to G . A homomorphism $\delta: \Delta^1(X; G) \rightarrow \Delta^2(X; G)$ is defined by $\delta\psi([v_0, v_1, v_2]) = \psi([v_0, v_1]) + \psi([v_1, v_2]) - \psi([v_0, v_2])$, a signed sum of the values of ψ on the three edges in the boundary of $[v_0, v_1, v_2]$, just as $\delta\varphi([v_0, v_1])$ for $\varphi \in \Delta^0(X; G)$ was a signed sum of the values of φ on the boundary of $[v_0, v_1]$. The two homomorphisms $\Delta^0(X; G) \xrightarrow{\delta} \Delta^1(X; G) \xrightarrow{\delta} \Delta^2(X; G)$ form a chain complex since for $\varphi \in \Delta^0(X; G)$ we have $\delta\delta\varphi = (\varphi(v_1) - \varphi(v_0)) + (\varphi(v_2) - \varphi(v_1)) - (\varphi(v_2) - \varphi(v_0)) = 0$. Extending this chain complex by 0's on each end, the resulting homology groups are by definition the cohomology groups $H^i(X; G)$.

The formula for the map $\delta: \Delta^1(X; G) \rightarrow \Delta^2(X; G)$ can be looked at from several different viewpoints. Perhaps the simplest is the observation that $\delta\psi = 0$ iff ψ satisfies the additivity property $\psi([v_0, v_2]) = \psi([v_0, v_1]) + \psi([v_1, v_2])$, where we think of the edge $[v_0, v_2]$ as the sum of the edges $[v_0, v_1]$ and $[v_1, v_2]$. Thus $\delta\psi$ measures the deviation of ψ from being additive.

From another point of view, $\delta\psi$ can be regarded as an obstruction to finding $\varphi \in \Delta^0(X; G)$ with $\psi = \delta\varphi$, for if $\psi = \delta\varphi$ then $\delta\psi = 0$ since $\delta\delta\varphi = 0$ as we saw above. We can think of $\delta\psi$ as a local obstruction to solving $\psi = \delta\varphi$ since it depends only on the values of ψ within individual 2-simplices of X . If this local obstruction vanishes, then ψ defines an element of $H^1(X; G)$ which is zero iff $\psi = \delta\varphi$ has an actual solution. This class in $H^1(X; G)$ is thus the global obstruction to solving $\psi = \delta\varphi$. This situation is similar to the calculus problem of determining whether a given vector field is the gradient vector field of some function. The local obstruction here is the vanishing of the curl of the vector field, and the global obstruction is the vanishing of all line integrals around closed loops in the domain of the vector field.

The condition $\delta\psi = 0$ has an interpretation of a more geometric nature when X is a surface and the group G is $\mathbb{Z}$ or $\mathbb{Z}_2$. Consider first the simpler case $G = \mathbb{Z}_2$. The condition $\delta\psi = 0$ means that the number of times that ψ takes the value 1 on the edges of each 2-simplex is even, either 0 or 2. This means we can associate to ψ a collection C_ψ of disjoint curves in X crossing the 1-skeleton transversely, such that the number of intersections of C_ψ with each edge is equal to the value of ψ on that edge. If $\psi = \delta\varphi$ for some φ , then the curves of C_ψ divide X into two regions X_0 and X_1 where the subscript indicates the value of φ on all vertices in the region.





The key to relating cohomology groups to homology groups is the observation that a function from i -simplices of X to G is equivalent to a homomorphism from the simplicial chain group $\Delta_i(X)$ to G . This is because $\Delta_i(X)$ is free abelian with basis the i -simplices of X , and a homomorphism with domain a free abelian group is uniquely determined by its values on basis elements, which can be assigned arbitrarily. Thus we have an identification of $\Delta^i(X; G)$ with the group $\text{Hom}(\Delta_i(X), G)$ of homomorphisms $\Delta_i(X) \rightarrow G$, which is called the *dual group* of $\Delta_i(X)$. There is also a simple relationship of duality between the homomorphism $\delta: \Delta^i(X; G) \rightarrow \Delta^{i+1}(X; G)$ and the boundary homomorphism $\partial: \Delta_{i+1}(X) \rightarrow \Delta_i(X)$. The general formula for δ is

$$\delta\varphi([v_0, \dots, v_{i+1}]) = \sum_j (-1)^j \varphi([v_0, \dots, \hat{v}_j, \dots, v_{i+1}])$$

Thus we have the algebraic problem of understanding the relationship between the homology groups of a chain complex and the homology groups of the dual complex obtained by applying the functor $C \mapsto \text{Hom}(C, G)$. This is the first topic of the chapter.

3.1 Cohomology Groups

Homology groups $H_n(X)$ are the result of a two-stage process: First one forms a chain complex $\cdots \rightarrow C_n \xrightarrow{\partial} C_{n-1} \rightarrow \cdots$ of singular, simplicial, or cellular chains, then one takes the homology groups of this chain complex, $\text{Ker } \partial / \text{Im } \partial$. To obtain the cohomology groups $H^n(X; G)$ we interpolate an intermediate step, replacing the chain groups C_n by the dual groups $\text{Hom}(C_n, G)$ and the boundary maps ∂ by their dual maps δ , before forming the cohomology groups $\text{Ker } \delta / \text{Im } \delta$. The plan for this section is first to sort out the algebra of this dualization process and show that the cohomology groups are determined algebraically by the homology groups, though in a somewhat subtle way. Then after this algebraic excursion we will define the cohomology groups of spaces and show that these satisfy basic properties very much like those for homology. The payoff for all this formal work will begin to be apparent in subsequent sections.

The Universal Coefficient Theorem

Let us begin with a simple example. Consider the chain complex

$$0 \longrightarrow \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0$$

$$\quad \quad \quad \parallel \quad \quad \parallel \quad \quad \parallel \quad \quad \parallel$$

$$\quad \quad \quad C_3 \quad \quad C_2 \quad \quad C_1 \quad \quad C_0$$

where $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ is the map $x \mapsto 2x$. If we dualize by taking $\text{Hom}(-, G)$ with $G = \mathbb{Z}$, we obtain the cochain complex

$$0 \longleftarrow \mathbb{Z} \xleftarrow{0} \mathbb{Z} \xleftarrow{2} \mathbb{Z} \xleftarrow{0} \mathbb{Z} \longleftarrow 0$$

$$\quad \quad \quad \parallel \quad \quad \parallel \quad \quad \parallel \quad \quad \parallel$$

$$\quad \quad \quad C_3^* \quad \quad C_2^* \quad \quad C_1^* \quad \quad C_0^*$$

In the original chain complex the homology groups are $\mathbb{Z}$'s in dimensions 0 and 3, together with a $\mathbb{Z}_2$ in dimension 1. The homology groups of the dual cochain complex, which are called cohomology groups to emphasize the dualization, are again $\mathbb{Z}$'s in dimensions 0 and 3, but the $\mathbb{Z}_2$ in the 1-dimensional homology of the original complex has shifted up a dimension to become a $\mathbb{Z}_2$ in 2-dimensional cohomology.

More generally, consider any chain complex of finitely generated free abelian groups. Such a chain complex always splits as the direct sum of elementary complexes of the forms $0 \rightarrow \mathbb{Z} \rightarrow 0$ and $0 \rightarrow \mathbb{Z} \xrightarrow{m} \mathbb{Z} \rightarrow 0$, according to Exercise 43 in §2.2. Applying $\text{Hom}(-, \mathbb{Z})$ to this direct sum of elementary complexes, we obtain the direct sum of the corresponding dual complexes $0 \leftarrow \mathbb{Z} \leftarrow 0$ and $0 \leftarrow \mathbb{Z} \xleftarrow{m} \mathbb{Z} \leftarrow 0$. Thus the cohomology groups are the same as the homology groups except that torsion is shifted up one dimension. We will see later in this section that the same relation between homology and cohomology holds whenever the homology groups are finitely generated, even when the chain groups are not finitely generated. It would also be quite easy to